Emergent Dimensionality Space:
Atemporal Space with Emergent Dimensionality

Maximilian Golubev

Independent Researcher · April 2026

Abstract

This paper introduces the concept of Emergent Dimensionality Space (EDS) — a theoretical knowledge representation environment in which dimensionality is a property of each individual point rather than a global characteristic of the space. Unlike traditional vector spaces with fixed dimensionality, EDS allows objects of fundamentally different structural complexity to coexist within a single system without forced normalisation. The key elements are formally defined: the structured point, the open GlobalDimensionSpace registry, the weighted Jaccard coefficient as a similarity measure, the hyperdimensionality mechanism, and the structural transition operator T(P → Q). The space admits local geometries (Euclidean, hyperbolic, and others). EDS is proposed as a foundation for novel cognitive architectures, associative memory systems with variable complexity, and explainable models of structural reasoning. A minimal reference implementation is provided, demonstrating structural similarity computation and transition analysis on a four-concept example. Experimental results confirm the formal properties of the model and reveal a non-trivial interaction between Condition C and geometric vector structure.

1. Introduction

1.1. Limitations of Fixed Dimensionality and Implicit Flow

Contemporary knowledge representation systems rely on vector spaces of fixed dimensionality. This holds for both dense embeddings (word2vec, GloVe, BERT) and Knowledge Graph structures. All objects are forced to exist in a single space of identical dimensionality — a constraint that is not technical but conceptual in nature, following directly from the definition of a vector space.

The consequences are non-trivial. Simple objects (“red”, “one”) and complex ones (“justice”, “causality”) occupy coordinates in the same space. Heterogeneous entities — numbers, events, relations, abstractions — compete for the same axes. Hierarchical structures fit poorly into flat spaces, which motivated the introduction of hyperbolic embeddings [Nickel & Kiela, 2017]. Nevertheless, hyperbolic space remains a fixed global geometry.

Beyond dimensionality, most computational architectures assume an implicit flow: time, order, sequence. Even in systems that claim parallelism, operations are linearly ordered at the implementation level. Cognitive science, by contrast, has long established that memory works differently: memories are retrieved associatively and atemporally, and their “dimensionality” varies substantially [Tulving, 1983; Shallice, 1988].

1.2. Motivation from Cognitive Science

Human episodic memory is not organised like a database with a fixed schema. The memory of a forest’s scent and the memory of a mathematical proof have fundamentally different

internal structures and different numbers of significant “dimensions”. Yet both are stored in the same system and can be associatively linked through shared features.

Baars’s Global Workspace Theory [Baars, 1988] and predictive coding [Friston, 2010] both suggest that cognitive architecture is fundamentally multi-level and heterogeneous. None of the known neural memory mechanisms presupposes global normalisation of features to a single dimensionality. Fixed dimensionality is an artefact of mathematical convenience, not a reflection of the true nature of knowledge.

1.3. The Key Inversion

In standard geometry, space is primary and points are secondary: a point is a position within a pre-given containing container. EDS inverts this relation: space is not given in advance; it emerges between points as the aggregate of their structural relations. Geometry is not fixed globally but determined locally from the structure of each pair of points.

It is precisely this fundamental inversion — from “space is primary” to “points are primary” — that the present paper addresses.

2. Core Principles

2.1. Space as an Emergent Phenomenon

EDS does not posit space as a containing container. Space is defined as the aggregate of relations between points arising from their structure. This approach differs fundamentally from standard metric spaces, where the metric is globally fixed. In EDS the metric is local: for each pair of points it is determined on the basis of structural similarity.

2.2. A Point as a Structured Object

A point in EDS is not a position in space but an object with internal structure. In standard embeddings “point” and “vector” are synonyms. In EDS, a vector is merely one possible representation; the primary entity is a structure of named dimensions with local weights.

2.3. Dimensions as Semantic Axes

A dimension in EDS is a named semantic unit with a representation vector and an abstraction level. A point holds only its own projection: active dimensions with local weights. This sparse representation faithfully reflects the real structural complexity of an object: simple concepts have 2–5 active dimensions, complex ones have 10–20 or more, which avoids the artificial “inflation” or “compression” of objects.

2.4. Atemporality and the Absence of Global Flow

EDS contains no built-in notion of time, global computation order, or predefined traversal direction. The transition operator $T(P \rightarrow Q)$ is not movement in time but a structural transformation: a compatibility check and a possible temporary expansion of the point.

“Emergent Dimensionality Space” means precisely this: the space does not presuppose a global time step or a global information flow. Operations occur locally and on demand — only when an associative link is established between two points. This fundamentally distinguishes EDS from architectures in which the processing order is rigidly determined by graph topology or a temporal axis.

One qualification: atemporality pertains to the semantic level. Any physical implementation presupposes at least a partial order of operations. EDS does not prohibit this; it merely asserts that such an order is not part of the knowledge model.

3. Formal Definition

3.1. A Point and Its Internal Structure

Formally, a point P in EDS is defined as a 4-tuple:

$$P = \langle \text{content}(P),\ D(P),\ W(P),\ g_P \rangle$$

where:

- content(P) — the semantic content of the point;
- $D(P) \subseteq GDS$ — a finite set of active dimensions;
- $W(P) : D(P) \to \mathbb{R}_+$ — a weight function;
- g_P — the local metric in the neighbourhood of P.

The component g_P determines the geometric behaviour of the point in its neighbourhood and allows different regions of the space to have different metric natures — Euclidean, hyperbolic, or otherwise. Without this component, all points would be forced to share a single global geometry.

Dimensionality of a point: $\dim(P) = |D(P)|$ — a non-negative integer that may change over time.

3.2. GlobalDimensionSpace and Abstraction Levels

GlobalDimensionSpace is an open registry of all dimensions in the system:

$$GDS = \{\ d_i = \langle name_i,\ vec_i,\ L_i \rangle \mid i \in \mathbb{N}\ \}$$

where name_i is a string identifier; $vec_i \in \mathbb{R}^n$ is a vector representation; $L_i \in \mathbb{N}_0$ is the abstraction level. The registry is open: new dimensions are added without restructuring existing constructs.

The abstraction level L(d) encodes the hierarchical nature of dimensions: sensory and concrete features have L = 0, abstract categories have L > 0. This parameter is used in the compatibility condition (§3.4).

3.3. Centroid and Structural Similarity

The centroid of a point P is the weighted mean of the vectors of its active dimensions:

$$\text{centroid}(P) = \frac{\Sigma_\{d \in D(P)\}\ W(P,d) \cdot vec(d)}{\Sigma_\{d \in D(P)\}\ W(P,d)}$$

The centroid serves as the geometric centre of meaning of the point and is used to evaluate the compatibility of new dimensions.

The structural similarity of two points P and Q is defined as the weighted Jaccard coefficient:

$$sim(P,Q) = \frac{\Sigma_\{d \in D(P) \cap D(Q)\}\ \min(\ W(P,d),\ W(Q,d)\ )}{\Sigma_\{d \in D(P) \cup D(Q)\}\ \max(\ W(P,d),\ W(Q,d)\ )}$$

Key properties:

- $sim(P,Q) \in [0,1]$; sim = 1 for structural identity, sim = 0 when there are no shared dimensions;
- the metric is defined for points of any dimensionality, including $\dim(P) \neq \dim(Q)$;
- invariant to weight scaling: normalisation is built in via max.

The weighted Jaccard coefficient is preferable to cosine similarity in the EDS context: cosine requires a shared space of fixed dimensionality for both objects. Jaccard works naturally with sparse sets of different dimensionality and penalises structural mismatch through the union in the denominator.

3.4. Dimension Compatibility Conditions

Two conditions determine whether a point P can accept a new dimension d from GDS:

Layer Condition (C): $L(d) \leq \max_{d' \in D(P)} L(d')$

A dimension is accepted only if its abstraction level does not exceed the maximum level of the point's existing dimensions. The maximum, not the average, is used: the presence of even one highly abstract dimension means the point is already "ready" to operate at that level. The average would be too conservative: a point with one abstract and nine concrete dimensions would have a low mean level and would reject new abstract dimensions despite being genuinely capable of them.

Geometry Condition (B): $\text{cos_sim}(\text{vec}(d), \text{centroid}(P)) \geq \theta(P)$

where $\theta(P) \in (0, 1]$ is the local semantic selectivity threshold of the point. The vector of the new dimension must be sufficiently close to the centroid of the point, i.e. semantically consistent with its current structure.

Compatibility: $\text{compatible}(d, P) \Leftrightarrow C(d,P) \wedge B(d,P)$

3.5. Hyperdimensionality

Hyperdimensionality is a temporary state of a point in which it acquires a trial dimension for carrying out a structural transition:

```
H(P) = P ⊕ {d_temp}  when compatible(d_temp, P)
```

After the transition completes:

- if d_temp is structurally useful — i.e. $\text{cos_sim}(\text{vec}(d_temp), \text{centroid}(P)) \geq \theta(P)$ after the transition — it is consolidated: $D(P) := D(P) \cup \{d_temp\}$, with weight initialised as $W(P, d_temp) := \text{cos_sim}(\text{vec}(d_temp), \text{centroid}(P))$;
- otherwise d_temp is removed and the point returns to its prior state.

The threshold $\theta(P)$ may adapt after each successful transition: if the point has systematically accepted all proposed dimensions, $\theta(P)$ is raised; if transitions were frequently blocked, the threshold is lowered. The specific adaptation scheme for $\theta(P)$ remains an open question for future implementations.

3.6. The Transition Operator T(P → Q)

Algorithm 1. Transition Operator T(P → Q):

```
Input: point P, point Q
δ ← D(Q)  D(P)  // structural deficit
if δ = ∅: return ok  // transition is free
for each d_temp ∈ δ:
    if compatible(d_temp, P):
        H(P) ← H(P) ∪ {d_temp}
if H(P) ⊇ δ: return ok  // transition succeeds
else: H(P) ← ∅, return fail  // rollback
```

Success criterion:

$$T(P \to Q) = ok \Leftrightarrow D(P) \cup H(P) \supseteq D(Q)$$

A transition is not movement in space. It is a structural transformation: point P is temporarily or permanently modified in order to become structurally compatible with Q. The “distance” between points is not metric but the structural deficit |δ|.

After a successful transition, the weights of consolidated dimensions are updated: for each new $d \in \delta$, the weight W(P, d) is initialised as the cosine similarity between vec(d) and the updated centroid(P). This ensures semantic consistency of the new weights with the already accumulated structure.

3.7. Local Geometries

There is no global geometry in EDS. Each point P carries a parameter g_P specifying the metric in its neighbourhood. Permitted types: Euclidean, hyperbolic, spherical, and any user-defined metric.

The choice of geometry is semantically motivated. Hyperbolic geometry is natural for hierarchical structures: in a taxonomy tree, the distance between the root and leaves grows exponentially fast, which precisely matches the properties of Poincaré space [Nickel & Kiela, 2017]. Euclidean geometry is suitable for metric features where continuous distances matter: colour, temperature, frequency. Mixed concepts — for instance “musical style” with hierarchical genres and continuous characteristics (tempo, key) — can have a composite geometry realised through several active dimensions with different g_P.

Transitioning between regions with different geometries requires additional coordination via hyperdimensionality: the point temporarily acquires a dimension specific to the target geometry. In this way EDS naturally accommodates different types of geometry within a single point.

3.8. Illustrative Example

To make the formal construction concrete, we trace a minimal example with four points representing the concepts “cat”, “animal”, “warmth”, and “purring”. Suppose the GlobalDimensionSpace contains five named dimensions with abstraction levels: fur (L=0), warm (L=0), living (L=1), category (L=1), sound (L=0). The four points are defined as follows:

```
P₁ = "cat":     D = {fur: 0.9, warm: 0.8, living: 0.7, category: 0.6}
P₂ = "animal":  D = {living: 0.9, category: 0.8}
P₃ = "warmth":  D = {warm: 0.95}
P₄ = "purring": D = {sound: 0.85, warm: 0.5}
```

Structural similarity. Computing $sim(P_1, P_2)$ via the weighted Jaccard coefficient: the intersection {living, category} contributes min(0.7, 0.9) + min(0.6, 0.8) = 0.7 + 0.6 = 1.3; the union {fur, warm, living, category} contributes max values 0.9 + 0.8 + 0.9 + 0.8 = 3.4. Therefore $sim(P_1, P_2) = 1.3 / 3.4 \approx 0.38$. By contrast, $sim(P_1, P_3) = \min(0.8, 0.95) / (0.9 + 0.95 + 0.7 + 0.6) = 0.8 / 3.15 \approx 0.25$, reflecting a weaker structural overlap. The pair (P_1, P_2) is structurally closer than (P_1, P_3), which matches the intuitive semantic proximity.

Transition $T(P_3 \to P_1)$. The structural deficit is $\delta = D(P_1)\ \ D(P_3)$ = {fur, living, category}. For each $d \in \delta$, compatibility is checked against the current centroid of P_3 (which points in the direction of “warm”). Suppose $\theta(P_3) = 0.4$. The dimension fur has a vector geometrically close to warm (tactile-sensory cluster): cos_sim(vec(fur), centroid(P_3)) = 0.55 ≥ 0.4 → accepted into $H(P_3)$. The dimension living is more abstract: cos_sim = 0.35 < 0.4 → rejected. The dimension category is similarly abstract: cos_sim = 0.30 < 0.4 → rejected. Since $H(P_3)$ = {fur} does not cover δ, the transition $T(P_3 \to P_1)$ returns fail. The point P_3 cannot structurally reach cat directly — it lacks the abstract dimensions. Instead, the chain $P_3 \to P_1$ via P_4 succeeds: $T(P_3 \to P_4)$ has δ = {sound}, cos_sim(vec(sound), centroid(P_3)) = 0.45 ≥ 0.4 →

accepted; P_3 acquires {sound} and the transition succeeds. The chain continues: $T(P_4 \to P_1)$ proceeds similarly through shared warm.

This example illustrates three properties simultaneously: (1) objects of different dimensionality ($\dim(P_3) = 1$, $\dim(P_1) = 4$) coexist without normalisation; (2) the structural deficit δ determines reachability, not a global metric; (3) indirect associative paths emerge naturally when direct transitions fail.

Note: the example in this section is theoretical. The reachability of transitions depends on the geometric arrangement of dimension vectors, which in a physical implementation are initialised (e.g. randomly or via pre-trained embeddings). Section 6 presents experimental results for a concrete initialisation.

4. Key Properties

4.1. Plasticity and Structural Growth

EDS is plastic: the dimensionality of a point is not fixed. Through hyperdimensionality, points can acquire new dimensions, increasing structural complexity. Knowledge does not merely accumulate — it grows in depth: simple concepts acquire new facets as associations are established. The openness of GDS ensures scalability without destructive interference with existing structures.

4.2. Faithful Representation of Heterogeneous Knowledge

A fundamental property of EDS is the ability to store objects of radically different complexity without normalisation. Consider an example: the concept “apple” may have 3–4 active dimensions (colour, shape, taste, category “fruit”), whereas the concept “justice” will require 12–15 dimensions: philosophical context, legal norms, emotional valence, causal relations, cultural conditioning, and others. In a standard vector space both objects must have identical dimensionality: “apple” receives hundreds of zero coordinates, while “justice” is compressed to the same dimensionality as a concrete fruit. In EDS each object is faithfully described by exactly the dimensions that are meaningful for it.

4.3. Emergent Flow

Although EDS is atemporal, flow can emerge as a sequence of successful transitions $T(P_1 \to P_2 \to \cdots \to P_{\square})$. Such a path is not a predefined trajectory but a chain of structural transformations, each locally justified. This corresponds to cognitive models of associative thinking [Anderson, 1983].

4.4. Comparison with Existing Approaches

Property	Vector Space	Hyperbolic Space	Simplicial Complex	Sheaf NN	EDS
Dimensionality	Global, fixed	Global, fixed	Topological	Fixed at init	Local, dynamic
Metric	Euclidean / cosine	Hyperbolic	Combinatorial	Sheaves (linear maps)	Weighted Jaccard
Adding objects	Fit into space	Fit into space	Add simplices	Extend graph	Can expand GDS

Property	Vector Space	Hyperbolic Space	Simplicial Complex	Sheaf NN	EDS
Heterogeneity	Problem	Partially solved	Not solved	Partially	Norm
Structural growth	Absent	Absent	Partially	Via training (gradient)	Via hyperdimensionality (dynamic)
Local geometry	No	No	No	No	Yes
Learning	Gradient descent	Gradient descent	No	Gradient descent	Not required (structural transitions)

Table 1. Comparison of EDS with leading approaches. Sheaf NN — Sheaf Neural Networks.

5. Related Work

5.1. Vector Embeddings

Dense vector representations — word2vec [Mikolov et al., 2013], GloVe [Pennington et al., 2014], BERT [Devlin et al., 2019] — constitute the dominant knowledge representation paradigm. The fundamental limitation is fixed dimensionality for all objects. Gaussian embeddings [Vilnis & McCallum, 2014] introduce a variable “spread” of the point, but the space itself remains globally fixed. In EDS, variable dimensionality is semantically motivated and is a characteristic of the point itself rather than a parameter of its distribution.

5.2. Hyperbolic Spaces

Poincaré Embeddings [Nickel & Kiela, 2017] demonstrated that hyperbolic space effectively represents hierarchical structures. EDS shares this motivation but goes further: instead of a global hyperbolic geometry, local geometries are introduced, defined pairwise. EDS generalises hyperbolic spaces as a special case.

5.3. Hyperdimensional Computing and Vector Symbolic Architectures

HDC [Kanerva, 2009] and VSA [Gayler, 2003] use spaces of fixed high dimensionality (~10,000) with binding and superposition operations. Similarities with EDS: semantic operations on vectors, associative retrieval. The key difference: in HDC the dimensionality is fixed and identical for all symbols. EDS goes further, abandoning global dimensionality entirely — compatibility is built locally through conditions C and B.

5.4. Sheaf Neural Networks and Topological Approaches

Sheaf Neural Networks [Hansen & Gebhart, 2020; Barbero et al., 2022] are conceptually the closest existing approach to EDS. Each graph node has its own local feature space; alignment is determined by sheaves — linear maps. This resonates with the idea of local geometries in EDS.

Key differences: first, in Sheaf NN local spaces are fixed at initialisation and trained via gradient descent; in EDS they change structurally through hyperdimensionality without gradient training. Second, a sheaf defines a linear map between spaces of fixed dimensionality; the operator T(P→Q) defines a non-linear compatibility condition through the

deficit δ. Third, Sheaf NN operate on graph learning tasks; EDS is positioned as a framework for cognitive architectures without gradient training.

Among topological approaches, simplicial complexes and TDA [Edelsbrunner & Harer, 2010] are the most closely related. Simplicial complexes encode multi-level structures, but dimensionality is set globally and the metric is combinatorial rather than semantic. EDS builds structure bottom-up through compatibility conditions.

5.5. Modern Hopfield Networks

Classical Hopfield networks [Hopfield, 1982] and Modern Hopfield Networks [Ramsauer et al., 2020] implement associative retrieval through a global energy function. Memory capacity is exponential in dimensionality; dynamics involve weighted summation with softmax.

Key differences from EDS: all patterns have identical fixed dimensionality; the energy function is global and differentiable; there is no structural growth of individual patterns; the retrieval threshold is not a local parameter. EDS proposes an alternative foundation: instead of global energy — local conditions C and B; instead of fixed patterns — points with mutable structure and local thresholds θ(P).

5.6. Symbolic and Neuro-Symbolic Approaches

In classical symbolic systems (Minsky frames, OWL ontologies) objects have explicit structure, but it is fixed by a schema. Neuro-symbolic architectures [Garcez et al., 2022] combine neural and symbolic representations, typically by distinguishing two levels. EDS proposes a unified environment in which structural flexibility and semantic meaningfulness coexist within a single formalism.

6. Implications and Potential Applications

6.1. Memory Systems with Variable Complexity

EDS naturally models semantic memory with heterogeneous structure. A system built on EDS does not require a pre-defined schema: new types of objects add their dimensions to GDS without disrupting existing structures. This opens the possibility of incremental learning in which knowledge is built upon rather than overwritten.

6.2. Structural Reasoning

The operator T(P→Q) provides an explicit mechanism for associative reasoning: each transition is explained through the deficit δ and the compatibility conditions. Analogy is formalised as a successful transition through hyperdimensionality. A metaphor is a transition with partial mismatch: H(P) ⊇ δ holds only partially, but the transition is established anyway with reduced weight.

6.3. Foundation for Cognitive Architectures

EDS is not a finished cognitive model but an environment for building one. On top of EDS one can implement associative networks with mutable connections, multi-level architectures with different geometries at different levels of abstraction, and reasoning systems with explicit traces of structural transformations. This corresponds to the philosophy of ACT-R [Anderson et al., 2004].

6.4. Philosophical Implications

EDS implicitly supports a relational ontology: objects exist through the dimensions they share with others. Dimensionality as a measure of complexity resonates with Tononi's concept of information integration (Φ, IIT) [Tononi, 2004]. The atemporality of EDS reflects the view that knowledge structures have no privileged direction of time.

7. Experiments

7.1 Setup

The reference implementation uses Python and NumPy. The GlobalDimensionSpace contains 12 named dimensions; dimension vectors are drawn from $\mathbb{R}^8$ and initialised randomly with seed = 42. Abstraction levels are assigned as follows:

```
fur(0), warm(0), sound(0), soft(0), fuzzy(0), quiet(0), loud(0)  —
concrete (L=0)
living(1), category(1), mammal(1), entity(1), feeling(1)        —
abstract (L=1)
```

The global threshold is θ = 0.3. The four points from §3.8 are used: `cat`, `animal`, `warmth`, `purring`. Code is available at: [https://github.com/lastnnilabs/eds-reference-implementation].

7.2 Structural Similarity

Pair	sim(P, Q)
cat / animal	0.38
cat / warmth	0.25
cat / purring	0.13
warmth / purring	0.28

Table 2. Pairwise structural similarity for the four example concepts.

The results align with the theoretical computations of §3.8. The pair cat / animal has the highest similarity (0.38), consistent with their shared abstract dimensions. cat / purring scores lowest (0.13), reflecting their sparse structural overlap under the concrete seed=42 initialisation.

7.3 Transition Analysis

Transition	Result
T(warmth → cat)	FAIL
T(warmth → purring)	OK
T(purring → cat)	FAIL
T(animal → cat)	OK

Table 3. Transition operator results under seed=42 initialisation.

7.4 Deep Analysis: T(purring → cat)

This transition is the central result of the section. The structural deficit is δ = {fur, living, category}. The state of purring has max_level = 0: both active dimensions (warm and sound) are concrete (L=0). The per-dimension analysis is as follows:

Dimension	Level	cos_sim	Cond. C	Cond. B	Result
category	1	+0.07	FAIL	FAIL	blocked
fur	0	−0.48	PASS	FAIL	blocked
living	1	+0.24	FAIL	FAIL	blocked

Table 4. Per-dimension compatibility analysis for T(purring → cat).

Two independent blocking mechanisms are at work:

- Condition C blocks living and category — a point without any abstract dimensions cannot accept abstract concepts.
- Condition B blocks fur — geometric anti-correlation with centroid(purring): cos_sim = −0.48.

This is not a defect of the model but a substantive property. The concept purring is semantically distant from cat as a category — a direct transition requires structural intermediaries.

7.5 Emergent Chain

The theoretical example in §3.8 assumes geometric proximity between fur and the centroid of purring. With seed=42 initialisation, cos_sim(fur, centroid(purring)) = −0.48, making this assumption false. This highlights a key property of EDS: transition reachability is not purely structural but depends on the geometric arrangement of dimension vectors. The chain $P_3 \to P_4 \to P_1$ *is theoretically valid but geometrically contingent.*

8. Discussion and Open Questions

8.1. Initialisation and Weight Learning

The reference implementation provided in Section 7 constitutes a first empirical validation. Full-scale validation on real knowledge bases remains future work. No procedure for learning weights W(P,d) from data has been defined. No method for initialising vec(d) for new dimensions has been specified. Both questions are essential for practical implementation.

8.2. Computational Complexity

Computing sim(P,Q) for a single pair of points has complexity $O(|D(P)| + |D(Q)|)$. For N points with mean dimensionality $\bar{d}$, the full similarity matrix requires $O(N^2 \cdot \bar{d})$ operations — infeasible for large systems. Practical implementation requires approximate nearest-neighbour (ANN) methods. Whether weighted Jaccard metrics admit efficient ANN indices comparable to LSH for Euclidean spaces remains open, though positive results are known for the unweighted variant [Shrivastava & Li, 2014].

8.3. Stability Under Unbounded Dimensional Growth

The hyperdimensionality mechanism does not bound the growth of dimensionality from above. Without an explicit criterion for the “usefulness” of a dimension, the system may “swell” and lose efficiency: accumulated redundant dimensions reduce semantic selectivity and slow the comparison operation. A forgetting mechanism is needed: reducing the weight of inactive dimensions with subsequent removal of those for which $W(P,d) < \varepsilon$. Analysis of the convergence of such a scheme is an open question.

8.4. Structural Energy and Convergence

Does there exist a global function analogous to energy in Hopfield networks? Tentatively, one may define the structural potential of a point:

$$E(P) = \Sigma_{Q : T(P \to Q) \text{ attempted}} |\delta(P,Q)| / |D(Q)|$$

— the mean fraction of incompatible dimensions across transitions from P. A simpler alternative: $E'(P) = 1 - \mathrm{sim}(P, P^*)$, where P^* is the "ideal" neighbour with maximum similarity. The question of monotone decrease of E(P) under iterative application of the operator requires separate analysis. Without a convergence guarantee the system may be unstable.

8.5. On Eliminating Implicit Order

The principle of atemporality raises the question: can implicit order be fully excluded from a computational implementation? In practice — no: any implementation presupposes at least a partial order of operations. EDS is atemporal in the sense of the absence of semantic time and a predefined control flow, but not in the sense of physical computation sequence. This distinction is important: semantic atemporality is a property of the model; operational sequence is a property of the implementation.

Acknowledging this boundary does not weaken the model; it makes it honest. EDS does not claim that computations occur "simultaneously" — it claims that the order in which they occur does not change the meaning of the result.

8.6. Possible Extensions

Promising directions include: (1) probabilistic weights $W(P,d) \in [0,1]$ with a fuzzy-set-theoretic interpretation; (2) a merge operation for two points as the structural union of active dimensions; (3) formalisation of forgetting through weight decay of inactive dimensions.

9. Conclusion

This paper has introduced the formal construction of the Emergent Dimensionality Space (EDS) — a knowledge representation environment with local, dynamic dimensionality. The central results are:

- a point as a structured object: the 4-tuple $\langle content(P), D(P), W(P), g_P \rangle$;
- the weighted Jaccard coefficient as a structural proximity measure for objects of any dimensionality;
- compatibility conditions C and B, formalising the structural growth of a point;
- the hyperdimensionality mechanism as the basis for dynamic change of dimensionality;
- the transition operator $T(P \to Q)$ via the structural deficit δ;
- local geometries as a consequence of abandoning global space;
- a reference implementation demonstrating that the formal properties hold under concrete vector initialisation, and revealing a non-trivial interaction between Condition C and vector geometry (Section 7).

EDS does not claim to replace existing approaches. Its value lies in the inversion of the fundamental assumption about the primacy of space, and in providing a formal language for modelling knowledge with fundamentally heterogeneous structure. The author hopes that the concept of Emergent Dimensionality Space will prove a useful conceptual foundation for the development of novel cognitive architectures, associative memory systems with variable complexity, and explainable models of structural reasoning.

Acknowledgements. The author thanks all those who discussed the ideas of this work. The work was carried out independently, without institutional affiliation.

References

[1] Anderson, J. R. (1983). A spreading activation theory of memory. Journal of Verbal Learning and Verbal Behavior, 22(3), 261–295.

[2] Anderson, J. R., Bothell, D., Byrne, M. D., Douglass, S., Lebiere, C., & Qin, Y. (2004). An integrated theory of the mind. Psychological Review, 111(4), 1036–1060.

[3] Baars, B. J. (1988). A Cognitive Theory of Consciousness. Cambridge University Press.

[4] Barbero, F., Bodnar, C., Velingker, A., Bronstein, M., & Veličković, P. (2022). Sheaf Neural Networks with Connection Laplacians. ICML Workshop.

[5] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL 2019.

[6] Edelsbrunner, H., & Harer, J. (2010). Computational Topology: An Introduction. AMS.

[7] Friston, K. (2010). The free-energy principle: a unified brain theory? Nature Reviews Neuroscience, 11(2), 127–138.

[8] Garcez, A. d'Avila, Gori, M., Lamb, L. C., Serafini, L., Spranger, M., & Tran, S. N. (2022). Neural-Symbolic Learning and Reasoning: A Survey and Interpretation. Neuro-Symbolic AI.

[9] Gayler, R. W. (2003). Vector Symbolic Architectures answer Jackendoff's challenges for cognitive neuroscience. ICCS/ASCS.

[10] Hansen, J., & Gebhart, T. (2020). Sheaf Neural Networks. NeurIPS 2020 Workshop TDA.

[11] Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. PNAS, 79(8), 2554–2558.

[12] Kanerva, P. (2009). Hyperdimensional Computing: An Introduction. Cognitive Computation, 1(2), 139–159.

[13] Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed Representations of Words and Phrases. NeurIPS 2013.

[14] Nickel, M., & Kiela, D. (2017). Poincaré Embeddings for Learning Hierarchical Representations. NeurIPS 2017.

[15] Pennington, J., Socher, R., & Manning, C. D. (2014). GloVe: Global Vectors for Word Representation. EMNLP 2014.

[16] Ramsauer, H., Schäfl, B., Lehner, J., Seidl, P., Widrich, M., Gruber, L., & Hochreiter, S. (2020). Hopfield Networks is All You Need. ICLR 2021.

[17] Shrivastava, A., & Li, P. (2014). Asymmetric LSH (ALSH) for sublinear time maximum inner product search. NeurIPS 2014.

[18] Tononi, G. (2004). An information integration theory of consciousness. BMC Neuroscience, 5, 42.

[19] Tulving, E. (1983). Elements of Episodic Memory. Oxford University Press.

[20] Vilnis, L., & McCallum, A. (2014). Word Representations via Gaussian Embedding. arXiv:1412.6623.

[21] Golubev, M. (2026). EDS reference implementation. GitHub. [https://github.com/lastnnilabs/eds-reference-implementation]

Glossary

Table 5. Summary glossary of EDS terms and notation.

Term / Symbol	Definition
EDS	Emergent Dimensionality Space — a knowledge representation environment with local, dynamic dimensionality.
Point P	Tuple ⟨content(P), D(P), W(P), g_P⟩ — the basic object of EDS.
GDS	GlobalDimensionSpace — an open registry of all named semantic dimensions in the system.
D(P)	Set of active dimensions of point P; |D(P)| = dim(P).
W(P, d)	Weight of dimension d in point P; W : D(P) → $\mathbb{R}_+$.
centroid(P)	Weighted mean of the active dimension vectors — the geometric center of meaning of the point.
sim(P, Q)	Weighted Jaccard coefficient over active dimensions; sim ∈ [0, 1].
Condition C	Abstraction-level constraint: L(d) ≤ max_{d' ∈ D(P)} L(d').
Condition B	Geometric constraint: cos_sim(vec(d), centroid(P)) ≥ θ(P).
θ(P)	Local geometric compatibility threshold of point P.
H(P)	Hyperdimensionality — temporary state of a point with a trial dimension d_temp.
δ	Structural deficit: δ = D(Q) D(P).
T(P → Q)	Transition operator; succeeds ⇔ D(P) ∪ H(P) ⊇ D(Q).
g_P	Local metric in the neighbourhood of point P.
L(d)	Abstraction level of dimension d; L ∈ $\mathbb{N}_0$.

Appendix A: Code Listing

The full source code of the reference implementation (demo.py) is available on GitHub at: [https://github.com/lastnnilabs/eds-reference-implementation]. The repository includes a README with installation instructions:

```
pip install numpy
python demo.py
```

The implementation covers: GDS initialisation with named dimensions and abstraction levels; centroid computation; weighted Jaccard similarity; compatibility checks (Conditions C and B); the transition operator T(P → Q) with rollback; and full logging of per-dimension analysis for blocked transitions.